EVIDENCE RELEASE · FIXED-ISSUED TURBIDITY

A bounded monitoring decision

Lower forecast error; weaker alert allocation.

1014

scored North windows

6.8%

relative log-error reduction

46

proposed monitoring slots

The primary error test passed. The frozen alert policy did not improve monitoring efficiency. Continue prospective validation before operational allocation.

NEXT 6 h MAXIMUM · 30 min ASSUMED LATENCY · 2023–2024

The earlier 48 h storm task is a separate experiment.

SOURCE · held-out North archive / fixed model · METHOD · paired calendar blocks
CONFIDENCE · primary test passed · USE · monitoring pilot; no causal treatment or ROI claim

Repair the task before adding complexity

Constant antecedent rain

Fixed issues; observed rain

Recent turbidity omitted

Anchor to recent turbidity

Future-defined storm start

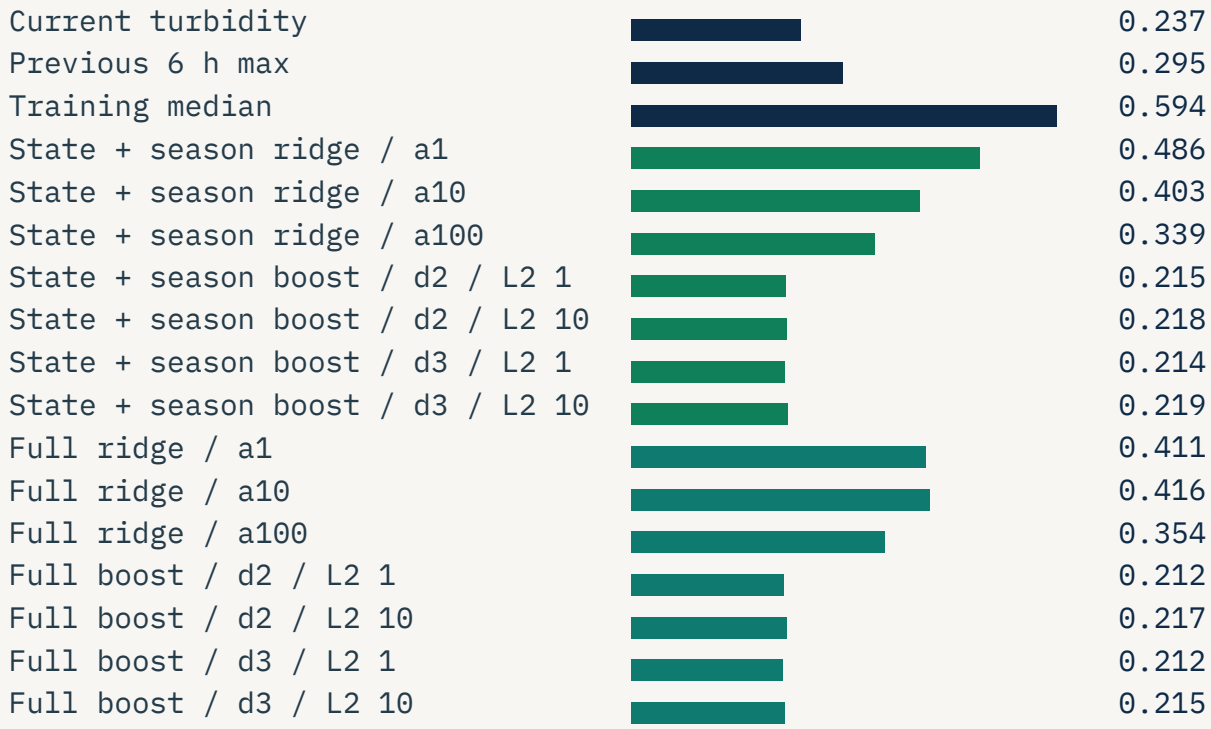
Past features; future target

Few storms and long maxima

More windows; purged time blocks

879 fitted windows · 8 folds

2021–2022 development only



SOURCE · job29758 OOF rows · METHOD · 7 day purge; 17 frozen candidates
CONFIDENCE · model-selection evidence only · USE · choose once, then freeze

UNTOUCHED NORTH · PRIMARY RESULT

Compare on the same accepted windows

POOLED ERROR IN log1p(FNU) · LOWER IS BETTER



GAIN OVER THE FROZEN BASELINE · LOG ERROR

0.0102 [0.0016, 0.0174]

Paired 95% interval; positive favors the learned model.

Year	n	Current	State	Full
2023	584	0.153	0.147	0.144
2024	430	0.143	0.139	0.131

Year summaries are secondary; no winner changes.

SOURCE · held-out2023/2024 North · METHOD · paired 7 day clusters, 2,000 draws
CONFIDENCE · interval reflects chosen block design · USE · conditional forecast skill

MONITORING POLICY · OPERATING CONSTRAINTS

Allocate before outcomes become known

Forecast	Chronological	Daily capacity	Outcome audit
Strict threshold	First crossing	One slot per day	Unknown stays counted

Monitoring counts	Current	Frozen model
All eligible forecasts	1021	1021
Accepted slots / all outcomes	38	46
Unknown outcomes	0	2
True alerts / known	26	25
False alerts / known	12	19
Missed high windows	54	55
Precision / recall	68.4% / 32.5%	56.8% / 31.2%

Candidate threshold uses ≤10% of development forecasts; test alert rate can differ.

HIGH TARGET DEFINITION

4.62 FNU

Development 90th percentile. A descriptive monitoring threshold, not a regulatory limit.

PLANNED COST ONLY

slots × lab cost

SOURCE · frozen thresholds and final allocations · METHOD · capacity before target QA
CONFIDENCE · retrospective decision counts · USE · plan sampling; no samples or ROI claimed

UNCERTAINTY · EMPIRICAL RESIDUAL BANDS

Measure coverage instead of promising it

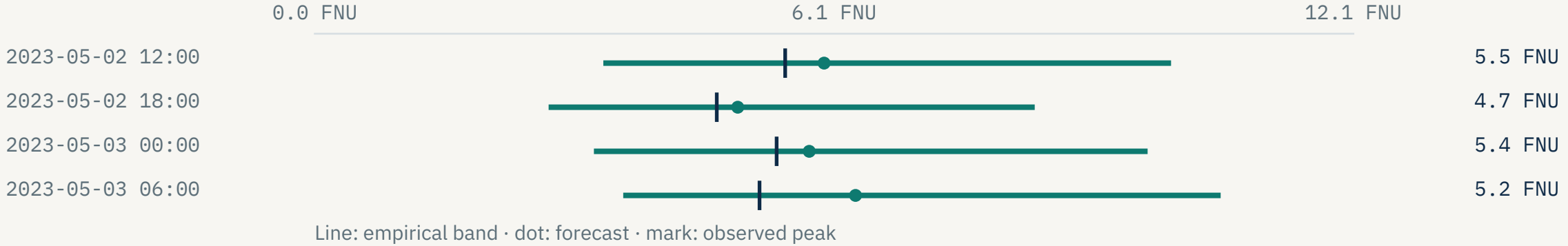
Band	All	2023	2024	Width(FNU)
Current	94.7%	95.4%	93.7%	3.8
State + season	94.7%	95.0%	94.2%	3.2
Full	94.9%	95.0%	94.7%	3.3

Nominal coverage: 90%. Radius learned only from purged development residuals.

Temporal drift can break calibration. No guaranteed coverage.

Full 95% CI: 93.2%–96.3%

FIRST FOUR SCORED WINDOWS · CHRONOLOGICAL RULE



Line: empirical band · dot: forecast · mark: observed peak

SOURCE · frozen OOF bands and final native maxima · METHOD · no final recalibration
CONFIDENCE · measured coverage only · USE · uncertainty disclosure, not a payout trigger

EXTERNAL EAST · BOUNDARIES AND PROVENANCE

North skill does not transfer to East

1481

scored East windows

0.095

state + season log error

0.080

current baseline log error

PAIRED GAIN · SAME NORTH-TRAINED STATE MODEL

`-0.0150 [-0.0268, -0.0040]`

East uses the frozen North state-plus-season model and North thresholds. It adds no local fitting, discharge or rainfall inputs.

WHAT WOULD CHANGE THE DECISION

Loss of skill, poor band coverage, missed high windows or insufficient source availability can stop a prospective pilot.

No causal treatment effect · no sediment mass · no routing validation
No avoided loss, ROI, nature-credit value or insurance payout established.

SOURCE · USGS North401530105475401 / East09013500 · model 6f5da1438b8c
METHOD · frozen temporal/external tests · CONFIDENCE · stated scope · USE · monitoring pilot